Multi-Modal Counterfeit Medicine Detection System

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***Abstract*—Counterfeit medicines are increasingly becoming a global threat to public health that mainly affects low- and middle-income countries (LMICs). They often look quite similar to legal pharmaceutical products and thus escape notice. Additionally, these can cause treatment failures and toxicological adverse effects, subsequently causing an increase in antimicrobial resistance. We present a multi-modal counterfeit drug detection system through recent works on multi-modal learning to mitigate these challenges. The system employs visual data along with textual data captured by optic character recognition or OCR for detecting non-conformance. Additionally, we present a refined MobileNetV2 architecture along with a GRU-based RNN designed for the effective processing of textual information. The framework also implemented sophisticated techniques for text augmentation and class balancing. Through the optimization of thresholds, the system achieves high recall along with a significant increase in accuracy for counterfeit medicine detection. The model is then integrated into a Flask web app with PaddleOCR for real-time output.**

***Index Terms*—Counterfeit Detection, Multi-Modal Learning, OCR, MobileNetV2, GRU, Deep Learning, Pharmaceutical AI, Real-Time Inference**

I. Introduction

One of the burning issues facing the global healthcare sector is the presence of counterfeit medicines. It especially targets vulnerable populations living in low and middle-income countries. The World Health Organization (WHO) has constantly issued warnings about the increase in the number of unacceptable and falsified medicines. It is estimated that over 10 percent of medical products in low and middle-income countries are fake [16].

In the past, it is seen that the government and pharmaceutical companies have used anti-counterfeit technologies such as holograms, barcodes, QR codes, and tamper-evident packaging. Even though these styles are effective to some extent, they can easily be copied by counterfeiters.

With the proliferation of smartphones, camera-equipped gadgets, and increased access to artificial intelligence (AI) technologies, there has been a shift toward software grounded fake discovery systems. These AI-driven results influence computer vision and machine learning to dissect packaging images, compare them with known authentic examples, and predict the liability of a product being fake. Still, numerous of these results concentrate on unimodal data either visual features or textual content — failing to subsidize on the synergistic power of combining both modalities.

Multi-Modal Counterfeit Medicine Detection System (MM-CMDS) is our proof-of-concept AI-powered framework that fuses visual cues from packaging with semantic information obtained through Optical Character Recognition (OCR). The system is designed to describe subtle inconsistencies that either the visual or textual modality might miss when considered in isolation.

MobileNetV2 serves as our image encoder due to its lightweight nature and strong performance on mobile devices. For textual data, we use a character-level *Gated Recurrent Unit (GRU)* network that processes OCR output into meaningful embeddings. The outputs of both branches are fused and passed through a dense classifier for final prediction.

In addition to presenting the architectural and training details, this paper demonstrates a practical deployment of MM-CMDS as a Flask web application integrated with PaddleOCR. This end-to-end system allows users to upload images and receive real-time predictions, complete with confidence scores and extracted text.

The rest of this paper is organized as follows. Section II discusses the prior works. Section III presents the problem statement. Section IV elaborates on the methodology. Section V covers data collection and preprocessing. Section VI details experimental results. Section VII concludes the paper.

II. Literature Review

The emergence of counterfeit drugs is one of the major problems in the world of health and economics. The range of research includes traditional packaging styles as well as modern machine learning techniques. This segment discusses anti-counterfeiting techniques based on computer vision, text recognition, and multi-modal techniques.

A. Traditional and Supply Chain-Based Techniques

Anti-counterfeiting techniques can be classified into three types: overt, covert, and supply chain-based techniques.

Overt techniques include holograms and special packaging schemes that enable consumers to check authenticity by sight [1]. While low-cost, overt features are increasingly vulnerable to duplication [2].

Covert techniques involve invisible labels such as UV inks or chemical taggants, offering better security but requiring special equipment for verification [3]. Track-and-trace systems enhance supply chain integrity through serialization and QR/barcode scanning, secured with blockchain technology [4], [5]. However, they provide limited help in offline scenarios [6].

B. Image-Based Detection by Deep Learning

Previous approaches have made use of handcrafted features such as SIFT [7] and machine learning classifiers such as SVM [8]. Deep learning is transforming this area. Deep models such as VGGNet [9], ResNet [10], and object detection models such as Faster R-CNN [11], SSD [12], and YOLO [13] have been explored for packaging verification. The accuracy-efficiency trade-off is enhanced by EfficientNet [14]. Recent studies such as Motwani et al. [15] used CNNs for classifying medicines. However, visual-only models may overlook subtle textual differences.

C. Text-Based Verification with OCR

Counterfeit packaging often contains textual errors [6], [16]. Printed text can be recognized using OCR tools like Tesseract [18], PP-OCR [19], and converted into machine-readable text. Modern OCR techniques involve the usage of CNN, RNN, and Transformers [17]. The accuracy is decreased if the background is blurry.

D. Multi-Modal Learning and Our Contribution

Multi-modal learning merges information from multiple sources [20], [21]. Our MM-CMDS uses MobileNetV2 [22] for visual features and a GRU-based encoder [23] for OCR text sequences. Features are concatenated and passed through a dense classifier trained with class-weighted cross-entropy loss and threshold tuning. Focal Loss [24] and threshold tuning enhance minority class detection.

E. Comparative Analysis

MM-CMDS fuses a MobileNetV2 image encoder with a character-level GRU on OCR text (PaddleOCR, Flask), trained on 4,071 images (3,951 authentic / 120 counterfeit), yielding test accuracy 87.50% with counterfeit precision 0.3333, recall 0.6000 and F1 0.4286. Image-only systems (CBS-YOLOv8 [26], mAP ≈97.4%) excel on narrow visual tasks; text-centric detectors (DLI-IT [27], TextSleuth [29]) show strong label retrieval and tamper localization. Applied pipelines (Edegbe & Tone [31]) combining logo detection, OCR and spell-check report logo mAP ≈ 79.3%; multimodal LLM approaches (Zakaria et al. [32]) achieve binary accuracy up to ≈74.6%. MM-CMDS sits as a practical multimodal prototype needing more counterfeit data, improved OCR robustness, and latency optimizations to be production-competitive.

III. Problem Statement

The problem of counterfeit drug detection involves analyzing packaging intentionally designed to mimic authentic products. Purely visual inspection is often insufficient. We formalize our problem as follows:

A. Objective

Given an image of a medicine's outer packaging, determine whether the product is classified as **'Authentic'** or **'Counterfeit'** based on both visual and textual features.

B. Inputs and Outputs

- **Input:** RGB image x_{img} of size $H \times W \times C$ and OCR text sequence $x_{\text{text}} = (c_1, c_2, \ldots, c_T)$ of length T.
- **Output:** Binary classification label $y \in \{0, 1\}$, where $0 =$ Counterfeit, $1 =$ Authentic.

C. Challenges

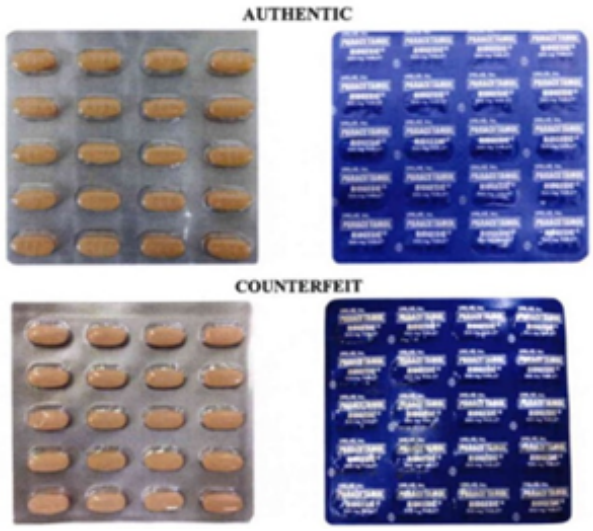

Fig. 1. Comparison of authentic and counterfeit drug packages highlighting subtle visual/textual differences. [25]

1) Visual Closeness: Counterfeiters replicate visual aspects such as colors, logos, fonts, and layouts.

2) Noisy OCR Data: Text may be distorted due to glare or complex fonts; OCR engines introduce errors.

3) Subtle Semantic Errors: Counterfeit packages contain near-correct text with minor typographical errors (e.g., "Paracetimol" vs "Paracetamol").

4) Class Imbalance: Authentic products dominate most datasets, requiring balancing via data augmentation or class-weighted loss.

5) Real-Time Requirements: Detection must be computationally efficient for real-time inference on low-power devices.

D. Approach

We propose a robust **multi-modal learning system** that simultaneously processes image and text modalities, supported by a lightweight, flexible deployment pipeline for real-world conditions.

IV. Methodology

MM-CMDS integrates both visual and textual information from medicine packaging. The architecture comprises three major components: a visual encoder, a text encoder, and a multi-modal fusion classifier.

A. Visual Encoder: MobileNetV2

The image modality is processed using MobileNetV2, a lightweight CNN optimized for mobile devices. Each input image $x_{\text{img}} \in \mathbb{R}^{224\times224\times3}$ is normalized and resized. We fine-tune from an intermediate block to extract domain-specific features.

$$f_{\text{img}} = \varphi_{\text{cnn}}(x_{\text{img}}) \tag{1}$$

where φ_{cnn} denotes the CNN encoder and $f_{\text{img}} \in \mathbb{R}^{d_1}$ is the resulting feature vector.

B. Text Encoder: Character-Level GRU

Text is extracted using PaddleOCR and converted to uppercase. It is tokenized at the character level and padded/truncated to fifty characters. The sequence is passed through an embedding layer followed by a Gated Recurrent Unit (GRU):

$$f_{\text{text}} = \varphi_{\text{rnn}}(x_{\text{text}}) \tag{2}$$

where φ_{rnn} represents the GRU network and $f_{\text{text}} \in \mathbb{R}^{d_2}$.

Text Augmentation Strategies:

- Character substitution (e.g., "O" ↔ "0", "I" ↔ "1")
- Random deletions and insertions
- Synthetic garbage text generation
- Full masking (empty input)

C. Multi-Modal Feature Fusion

The image and text features are concatenated and passed through a dense layer with ReLU activation:

$$f_{\text{fused}} = \text{ReLU}\left(W_f\left[f_{\text{img}}; f_{\text{text}}\right] + b_f\right) \tag{3}$$

D. Classification Layer

The fused features are passed through a softmax classifier to output class probabilities:

$$\hat{y} = \text{softmax}\left(W_o f_{\text{fused}} + b_o\right) \tag{4}$$

Here, $\hat{y} \in \mathbb{R}^2$ represents probabilities for the "Counterfeit" and "Authentic" classes.

E. Model Architecture Code Snippet

Listing 1.1. Model Fusion Architecture

```
# Image branch
image_input = Input(shape=(224, 224, 3))
base_model = MobileNetV2(include_top=False,
                         pooling='avg',
                         weights='imagenet')
image_feat = base_model(image_input)

# Text branch
text_input = Input(shape=(50,))
embed = Embedding(vocab_size, 32)(text_input)
gru_out = GRU(64)(embed)

# Fusion
fused = Concatenate()([image_feat, gru_out])
fused = Dense(128, activation='relu')(fused)
output = Dense(2, activation='softmax')(fused)

model = Model(inputs=[image_input, text_input],
              outputs=output)
```

F. Loss Function and Class Weighting

Due to severe class imbalance, we apply weighted categorical cross-entropy loss:

$$\mathscr{L} = -\sum_c w_c \cdot y_c \cdot \log(\hat{y}_c), \quad w_c = \frac{N}{2N_c} \tag{5}$$

- N: total number of samples
- N_c: number of samples in class c
- y_c and $\hat{y}_c$: true and predicted probabilities for class c

G. Threshold Optimization

The decision threshold τ is tuned empirically. If $\hat{y}_{\text{counterfeit}} \geq \tau$, classify as Counterfeit. Empirically, $\tau = 0.40$ provided the best balance of recall and precision.

H. Threshold Logic Snippet

Listing 1.2. Applying Threshold for Prediction

```
TAU = 0.40

def predict_with_threshold(model, image, text,
                           threshold=TAU):
    probs = model.predict([image, text])
    counterfeit_prob = probs[0][0]  # class 0
    if counterfeit_prob >= threshold:
        return "Counterfeit", counterfeit_prob
    else:
        return "Authentic", 1 - counterfeit_prob
```

V. Data Collection and Preprocessing

We have acquired 4,071 labeled samples—3,951 authentic and 120 counterfeit samples. Authentic images were gathered from verified pharmaceutical sources, while counterfeit samples were sourced from regulatory reports and manual annotations.

A. Image Preprocessing

- Resized to 224×224.
- Normalized using ImageNet statistics.
- Augmentations: rotation, scaling, flipping, brightness/-contrast shift, Gaussian blur.

B. Text Preprocessing

- Extracted using PaddleOCR.
- Converted to uppercase.
- Padded/truncated to fifty characters.
- Augmentations: character swaps, deletions, insertions, garbage replacement.

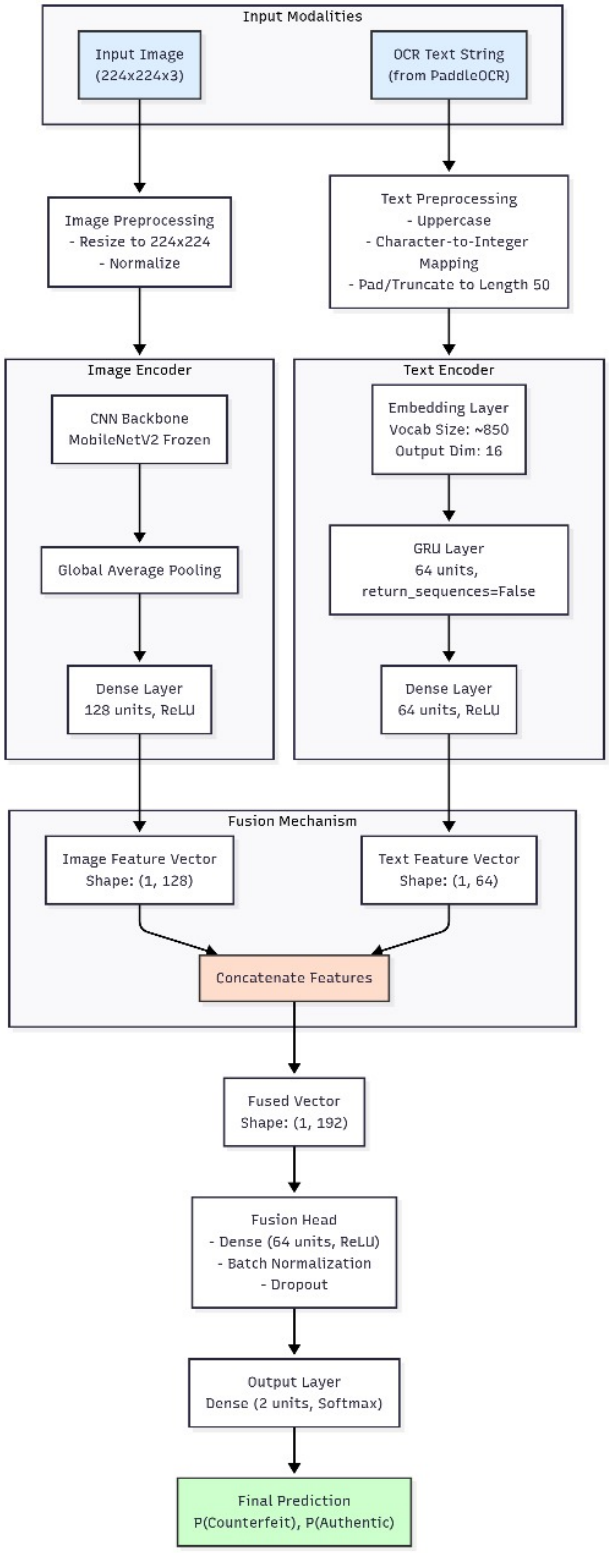

Fig. 2. MM-CMDS architecture: MobileNetV2 processes image data, GRU processes OCR text, and both streams are fused through a softmax classifier.

C. Class Balancing

1) Use of weighted loss as described in Equation (5).
2) Oversampling of counterfeit samples in each training batch.

These strategies ensure the model detects subtle patterns even under noisy or imbalanced data conditions.

VI. Evaluation

The performance of the MM-CMDS system was evaluated to determine the answers to three important queries:

1) How good is the performance of the model in detecting counterfeit medicine in unseen images?
2) How good is the performance of the model in generalizing different types of counterfeit medicine?
3) How good is the performance of the model when the quality of the image and/or the OCR is poor?

A. Evaluation Metrics

- **Accuracy:** What percentage of the predictions made by the model are actually correct?
- **Precision:** Out of all the predictions made by the model for being counterfeit, what percentage of them are actually counterfeit?
- **Recall:** Out of all the actual counterfeit medicines, what percentage of them are predicted by the model to be counterfeit?
- **F1-Score:** Harmonic mean of Precision and Recall.
- **ROC-AUC:** Measures class separation across different thresholds.

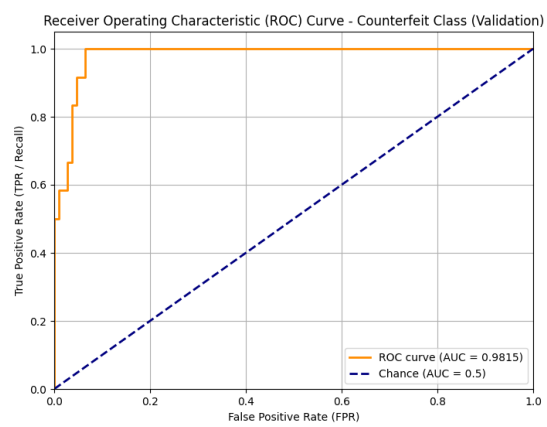

Fig. 3. ROC curve and threshold comparison for the counterfeit detection model.

B. Threshold Tuning and Baseline Comparison

The default value of 0.5 resulted in a model that was not as accurate in detecting counterfeits because of class imbalance. We tested different values for the threshold to enhance detection of counterfeits without raising unnecessary alarms.

C. Key Findings from Threshold Tuning

a) At Threshold 0.40 (Best F1-Score with Recall $\geq$ 0.90)::

- Precision: 0.6316
- Recall: 1.0000 (all counterfeits detected)
- F1-Score: 0.7742
- Accuracy: 94.17%

b) At Threshold 0.75 (Highest Precision with Recall $\geq$ 0.80)::

- Precision: 0.6667
- Recall: 0.8333
- F1-Score: 0.7407
- Accuracy: 94.17%

D. Performance on Test Data

- **Precision (Counterfeit class):** 0.3333
- **Recall (Counterfeit class):** 0.6000
- **F1-Score (Counterfeit class):** 0.4286
- **Overall Accuracy:** 87.50%
- **Weighted Average F1-Score:** 0.8907

E. Confusion Matrix

TABLE I
CONFUSION MATRIX FOR MM-CMDS ON TEST DATA

	Pred: Counterfeit	Pred: Authentic
Actual: Counterfeit	3	2
Actual: Authentic	6	53

- Three counterfeit samples were correctly predicted.
- Two counterfeit samples were missed (classified as authentic).
- Six authentic samples were incorrectly flagged as counterfeit.
- Fifty-three authentic samples were correctly identified.

Interpretation: The model is good at genuine and authentic product identification. With threshold tuning, the MM-CMDS can be made even more conservative so that no fake product is left out. This is important in public health.

Summary: The performance on real-world data is strong, especially with tuning on thresholds to minimize false negatives. This is an important trade-off for safety-critical use cases.

VII. CONCLUSION & FUTURE WORKS

In this paper, we introduced MM-CMDS, a Multi-Modal Counterfeit Medicine Detection System designed to identify falsified pharmaceutical packaging using visual and textual features. By integrating MobileNetV2 visual representations with character-level GRU textual embeddings from OCR outputs, the system captures both aesthetic discrepancies and semantic inconsistencies.

MM-CMDS achieves overall accuracy of **87.50%**, recall of **1.0000** for counterfeit on the validation set (at $\tau = 0.40$), and F1-score of **0.7742** under threshold tuning. These results further highlight the efficacy of multi-modal fusion and noise-based training.

Fig. 4. Flask UI for (top) Authentic, (below) Counterfeit predictions. [25], [34]

The MM-CMDS has been successfully deployed as a full-stack proof of concept using Flask and PaddleOCR, which makes it more applicable in the real world for use by pharmacists and regulators. It has shown promise in terms of inference time on regular hardware.

Future work includes: addressing counterfeit dataset scarcity; integrating detailed **pill and blister analysis** for physical feature verification; establishing a **user feedback loop** for continuous model improvement; and scaling to a broader production-ready framework.

The fusion of multiple modalities for counterfeit detection opens the door to many future AI applications in public health and safety domains. We envision MM-CMDS as a foundational framework for a broader scalable solution.

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